

# B and T Cell Receptors and Their Antigens

Antibody/TCR structure, V(D)J recombination, immunoglobulin classes, class switching, and antigen binding

## If you remember nothing else

- Know the overall structure of B-cell receptors/antibodies and T-cell receptors.
- Antibody Fab regions bind antigen; Fc regions drive many downstream effector functions.
- V(D)J recombination is a central mechanism for generating BCR and TCR diversity.
- Class switching changes the immunoglobulin heavy-chain constant region while preserving antigen specificity.

## Structural facts

- Papain and pepsin digestion are used to test how antibody fragments separate antigen-binding and Fc functions.
- Immunoglobulin class and subclass depend on the heavy-chain constant region.
- Antigen binds through complementarity-determining regions (CDRs).

## Rearrangement + switching

- Track the developmental stage together with the molecular event occurring at that stage.
- The lecture emphasizes the immunoglobulin heavy-chain sequence and the logic of class switching.
- TRECs and RAG-related disorders are used to connect receptor rearrangement with clinical immunodeficiency.

## Clinical recognition

- SCID/Omenn examples point back to failed receptor-generation biology.
- Hyper-IgM patterns point back to failed class switching.
- Molecular mimicry is used to connect antigen recognition with unintended cross-reactivity.